Optimal Stopping in High-Frequency Trading

Owen Den Uyl, Jenny Wang, Ken Bunworasate, and Xinze Li

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Abstract

Limit order books (LOBs) are often seen in high-frequency trading and consist of bids and asks at various different price levels. LOBs consist of limit buy and sell orders posted at a finite number of equidistant price levels on a given stock. When looking at maximizing profits in optimal execution in a high-frequency regime, we must take into account all the different outcomes that we have by the terminal time. We can either make a profit by posting a limit sell order and having it filled, take a loss by selling at the current best bid price, or take on an even larger loss if the best bid price drops. These features turn the profit maximization problem into an optimal stopping problem. We aim to determine exactly when the best time is to cancel our LO and post an MO to maximize expected profit.

1 Introduction

1.1 Limit Order Books

A limit order book (LOB), shown in Figure 1, is a collection of discrete equidistant price levels and volumes, representing prices at which investors are willing to buy or sell a certain amount of stock. The LOB's smallest possible step in price level is referred to as its tick size, which is usually \$0.01 for US stocks.

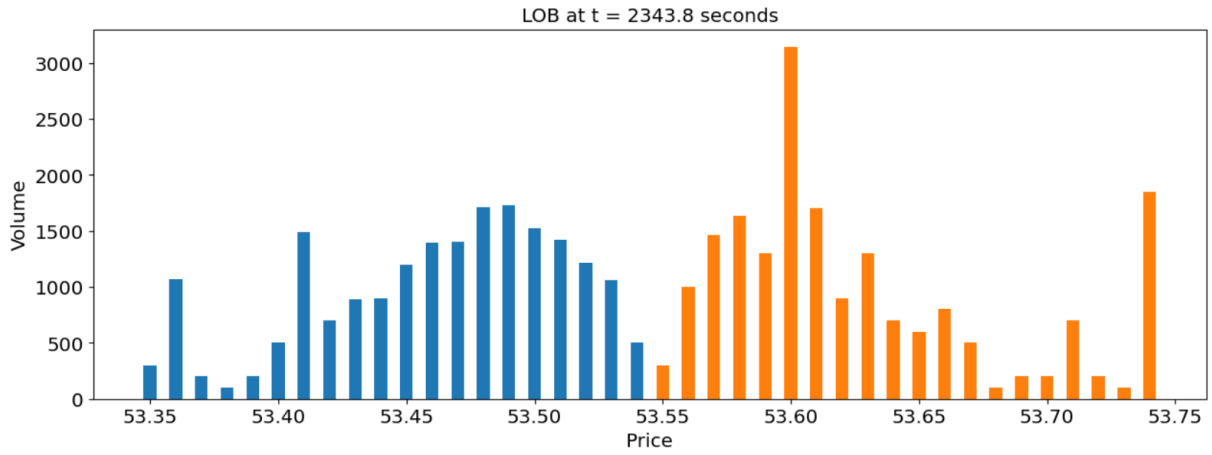


Figure 1: Example Limit Order Book for EBAY stock (November 6, 2014)

Investors can post limit orders (LOs), which have a price, location, volume, and type. Each of these limit orders is either a buy or sell order, and they provide liquidity in the market. Investors posting buy orders are looking to buy the stock at a certain price and volume, and those posting sell orders are looking to sell the stock at a certain price and volume. By the nature of these order

types, the lowest sell order must have a higher price than the highest buy order, as otherwise a transaction would just be made.

We refer to the largest price level at which there is a non-zero volume of limit buy orders as the best bid price and the smallest price level at which there is a non-zero volume of limit sell orders as the best ask price. The midprice, P, is defined as the midpoint between the best bid and ask prices, and the spread, s, is defined as the difference between the best bid and ask prices. As shown in Figure 2, the best bid and ask prices change quite frequently.

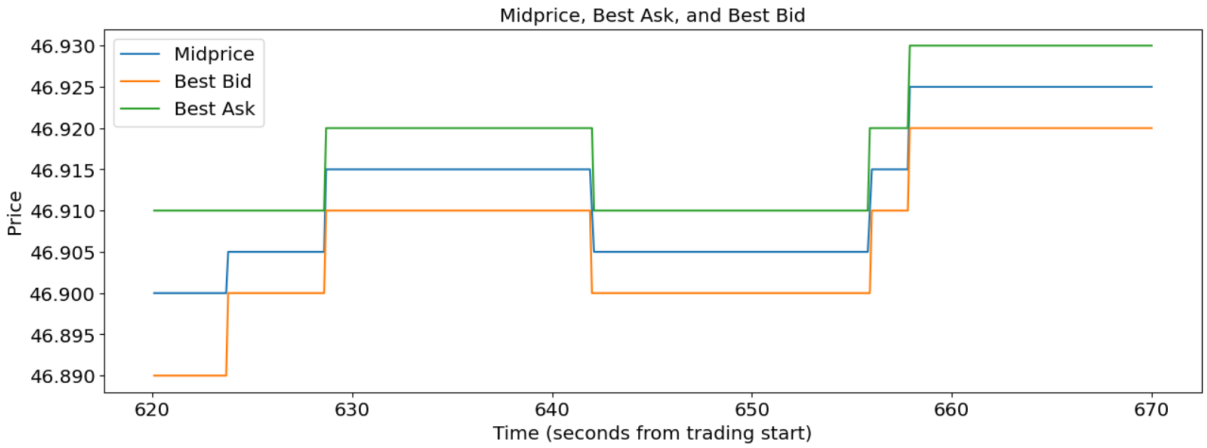


Figure 2: Best bid/ask prices and midprice for MSFT stock (November 3, 2014)

Market orders (MOs) fill/execute limit orders, consuming liquidity. MOs have a volume and type. When an MO arrives at a given price level and volume, the LOs at that price are executed according to a first-in-first-out principle. If a limit sell order happens to arrive at a price level lower than the best bid price (due to latency or other circumstances), it is instantly executed as an MO.

There are three main events that can occur in an LOB: an LO is added to the book, (partially) filled by an MO, or (partially) canceled. Variations in prices and volumes posted in an LOB exist due to both latency and a desire for queue priority if and when the posted price level becomes the best bid or ask price.

The microprice is the preferred metric of stock price in high frequency trading, as it reflects LO activity relatively well. We define the microprice as

$$P_M := A\frac{V^b}{V^a + V^b} + B\frac{V^a}{V^a + V^b},$$

where A and B are the best ask and bid prices, and V^a and V^b are the volumes (number of shares) of LOs posted at the best ask and bid prices respectively. Microprice has the advantage of taking on values that are not limited to integer multiples of tick size, allowing for more standard statistical analysis. The microprice also changes as frequently as the best bid and ask prices and volumes update.

1.2 Limit Order Imbalance

We define LO imbalance as

$$I := \frac{V^b - V^a}{V^b + V^a}.$$

The intuition behind imbalance is straightforward: when there is substantially more volume posted at the best bid than at the best ask, buying pressure dominates selling pressure. In these situations, the bid side, with its high volume, is more resilient to incoming sell orders, while the ask side is more vulnerable to depletion, increasing the likelihood of the price moving upward. Conversely, strongly negative imbalance indicates selling pressure and increases the probability of the price moving downward.

Note that I takes on values in the range $[-1, 1]$, where I is closer to -1 if $V^a >> V^b$, and I is closer to 1 if $V^b >> V^a$. We also observe that microprice can be expressed in terms of LO imbalance:

$$P_M = P + I\frac{A - B}{2},$$

where P is the midprice. LO imbalance works well as a predictor of market activity. Imbalance usually sits around zero, but trading becomes more common as it reaches either of its extremes. This motivates our use of imbalance as a finite-state signal. More broadly, Markovian and queue-reactive descriptions of the LOB are standard tools for connecting order-flow dynamics to short-horizon price and execution risk. Our empirical specification below follows this spirit, while estimating the transition law directly from observed imbalance data.

1.3 The Optimal Stopping Problem

Suppose we take on the role of an agent hired by a firm to sell some given amount of an asset (say, 100 shares of a stock) for the maximum expected profit possible. First, we post a sell LO of the given size, at the best ask price. Then, we wait until either the LO is filled or we decide to change it into a MO. We do not have access to any future data, so we must make a decision on when to change the LO into MO based on the information available at the time the decision is made. In our first model, we make a few simplifying assumptions:

- Until the process is over, the best ask and the best bid prices have spread equal to one tick size.
- We have the highest priority if an MO arrives and said MO will fill our whole LO.
- If best bid price drops, it drops by one tick size and we must immediately cancel our LO and submit an MO at the new price level.

The above assumptions allow us to model our behavior as the agent. At the initial time, we submit an LO at the best ask price. The agent now faces a trade-off. Leaving the limit order in the book creates the possibility of earning the spread if the order is executed, but also comes with the risk of losing money due to adverse price movements. Submitting a market order guarantees immediate execution but sacrifices the spread. The optimal stopping problem consists of determining the time at which the expected benefit of waiting no longer outweighs the execution risk.

We consider three times: the time at which the best bid price decreases (τ), the time at which the agent's LO is filled (σ), and the terminal time (T). The execution process ends at the minimum of these three times.

Let P_0 and Q_0 represent the midprice and quantity of shares being sold:

- If the bid price decreases before execution, the spread remains one tick and the new best bid becomes $P_0 - \frac{s}{2} - s = P_0 - \frac{3s}{2}$. The resulting revenue is thus $Q_0(P_0 - \frac{3s}{2})$.

- If the LO is filled before the terminal time, the agent makes a profit of $\frac{s}{2}$. The resulting revenue is thus $Q_0(P_0 + \frac{s}{2})$.
- If the terminal time is reached before the bid drops or the LO is filled, we sell at the best bid price for a revenue of $Q_0(P_0 - \frac{s}{2})$.

If no strategy is used in switching from LO to MO before S, the agent receives

$$X_T = \begin{cases} P_0Q_0 - 3sQ_0/2 & \text{if } \tau \leq (T \wedge \sigma) \\ P_0Q_0 + sQ_0/2 & \text{if } \sigma \leq (T \wedge \tau) \\ P_0Q_0 - sQ_0/2 & \text{if } T < (\tau \wedge \sigma) \end{cases}$$

We can turn this problem into an optimal stopping problem by finding some (random) time $\nu \leq T$ at which the agent cancels the LO and submits an MO that minimizes the probability of the first scenario. With strategy ν, the agent receives

$$X_T^\nu = \begin{cases} P_0Q_0 - 3sQ_0/2 & \text{if } \tau \leq (\nu \wedge \sigma) \\ P_0Q_0 + sQ_0/2 & \text{if } \sigma \leq (\nu \wedge \tau) \\ P_0Q_0 - sQ_0/2 & \text{if } \nu < (\tau \wedge \sigma) \end{cases}$$

The agent's overall goal can be thought of as solving

$$\max_{\nu \leq T}[\mathbb{E}(X_T^\nu)] = \max_{\nu \leq T}[(P_0Q_0 - 3sQ_0/2)\mathbb{P}(E_\tau) + (P_0Q_0 + sQ_0/2)\mathbb{P}(E_\sigma) + (P_0Q_0 - sQ_0/2)\mathbb{P}(E_\nu)],$$

where E_τ, E_σ, and E_ν represent the events where τ, σ, and ν are the minimum time, respectively.

2 The Initial Model

2.1 Parameters to Estimate

Our initial model uses only the LO imbalance as a stopping signal. We must take the maximum expected profit over all stopping times ν in $\{0, 1, ..., T\}$. We have two important probabilities to estimate:

- p_t – the probability that P moves down in the time interval $(t, t+1)$, given the current imbalance level, I_t
- π_t – the probability that the agent's LO is filled in the time interval $(t, t+1)$, given the current imbalance level, I_t

In our data, t is measured in tenths of a second, so the interval $(t, t+1)$ has length 0.1 seconds.

As the model will use LO imbalance as the predictive factor, we partition the interval $[-1, 1]$ into a grid X of $k+1$ equidistant points:

$$X = \{x_0, \ldots, x_k\}, \qquad x_i = -1 + \frac{2i}{k}.$$

For each time stamp, the observed imbalance value is mapped to the nearest point on the grid. We initially made the assumption that if the imbalance at time t, I_t, is equal to x_i, then I_{t+1} must be either x_{i-1}, x_i, or x_{i+1}, where each successive value of imbalance could be modeled by a mean-reverting random walk.

Our data, which includes full-day information for various large stocks, shows that the LO imbalance is not uniformly distributed. The imbalance values tend to resemble a bell-shaped curve, centered at the mean, as seen in Figure 3.

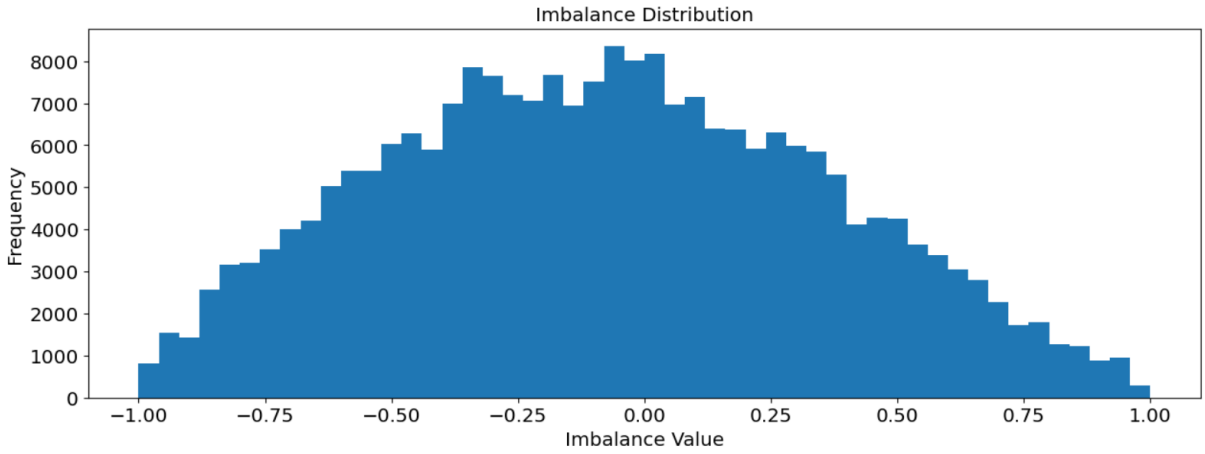


Figure 3: Distribution of imbalance values for MSFT stock (November 3, 2014)

Based on this bias toward the center, we decided to estimate two different probabilities for each imbalance state x, $q_{up}(x)$ and $q_{down}(x)$, representing the probabilities of the imbalance moving up or down in state. It follows that the probability of the imbalance remaining unchanged is equal to $1 - q_{up}(x) - q_{down}(x)$.

We now have a defined random walk for our imbalance states: given $I_t = x_i$,

$$I_{t+1} = \begin{cases} x_{i+1}, & \text{with probability } q_{up}(x_i), \\ x_{i-1}, & \text{with probability } q_{down}(x_i), \\ x_i, & \text{with probability } 1 - q_{up}(x_i) - q_{down}(x_i), \end{cases}$$

where q_{up} and q_{down} must be estimated for each imbalance state x_i.

2.2 Empirical Probabilities

We used the data we had with sufficient sample sizes (see Appendix section 4.1) to estimate q_{up}, q_{down}, p, and π for each of our imbalance states. As expected, q_{up} decreases and q_{down} increases with respect to imbalance level, shown in Figure 4. We also see that the probabilities are relatively small compared to the likelihood of the state not changing, which makes sense, as most of the time there will not be any market orders arriving in a given 0.1 second interval.

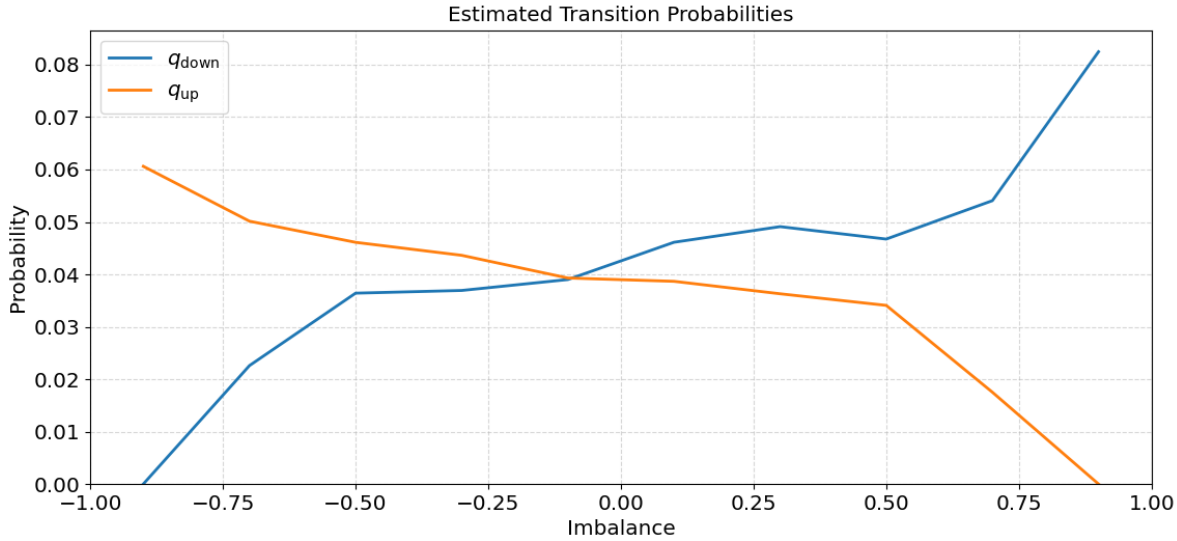


Figure 4: Transition probabilities, estimated from data on MSFT stock from November 4, 2014.

Similarly, we looked at p and π, the probabilities of the best bid dropping and the LO being filled, respectively. The results, seen in Figure 5, show that p decreases with imbalance and π increases with imbalance. These results make sense, as low imbalance indicates low volume at the best bid, typically driving price downward, and high imbalance indicates a high volume at the best bid and a low volume at the best ask, driving the price upward.

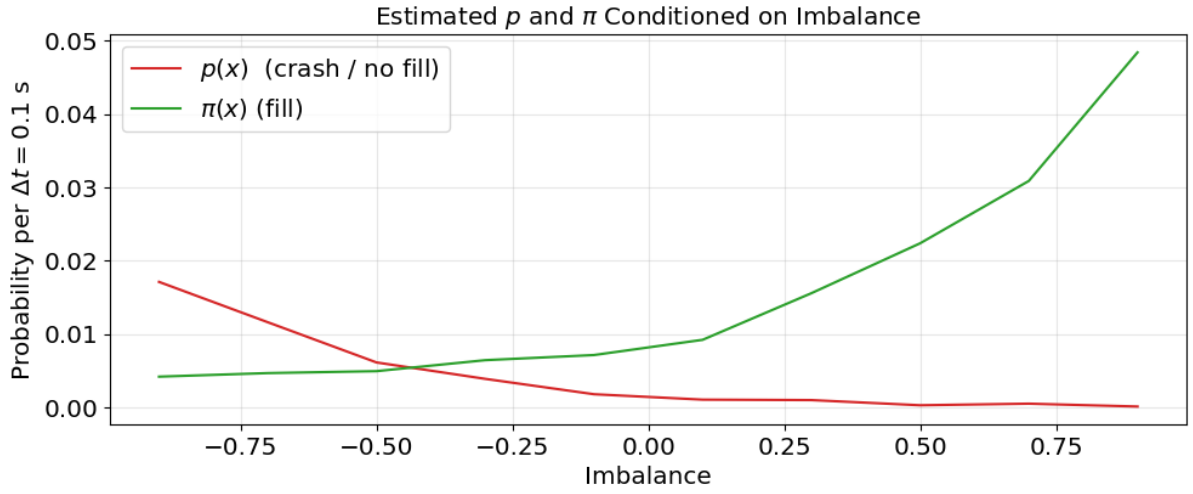


Figure 5: Empirically estimated p and π based on imbalance levels for MSFT stock on November 4, 2014.

2.3 The Value Function

Our state process is defined by $(I(t,x), \tau(t,x), \sigma(t,x))$. Then, the conditional value function at time t, given that the execution process has not terminated by time t and that $I_t = x$, known as the continuation function, is

$$V(t,x) = \max_{t \le \nu \le T} \left[(P_0 Q_0 - \tfrac{3sQ_0}{2}) \, \mathbb{P}(\tau(t,x) \le (\nu \wedge \sigma(t,x))) \right.$$

$$+(P_0Q_0 + \tfrac{sQ_0}{2})\,\mathbb{P}(\sigma(t,x) \le (\nu \wedge \tau(t,x)))$$
$$+(P_0Q_0 - \tfrac{sQ_0}{2})\,\mathbb{P}(\nu < (\tau(t,x) \wedge \sigma(t,x)))\Big].$$

At time $t = T$, if the execution process has not yet terminated, we must submit a market order. Thus,

$$V(T,x) = P_0Q_0 - \frac{sQ_0}{2}, \qquad \forall x.$$

The recursive dynamic programming principle (DPP) equation for the optimal stopping problem is

$$V(t,x_i) = \max\Bigg[P_0Q_0 - \frac{sQ_0}{2},$$

$$(P_0Q_0 - \tfrac{3sQ_0}{2})p(x_i) + (P_0Q_0 + \tfrac{sQ_0}{2})\pi(x_i) + (1 - p(x_i) - \pi(x_i))\mathbb{E}[V(t+1, I_{t+1}(t,x_i))]\Bigg],$$

for $i = 1, \ldots, k$ and $t = T-1, \ldots, 0$.

Given the dynamics of the imbalance process, we obtain

$$\mathbb{E}[V(t+1, I_{t+1}(t,x_i))] = q_{up}(x_i)V(t{+}1, x_{i+1}){+}(1{-}q_{up}(x_i){-}q_{down}(x_i))V(t{+}1,x_i){+}q_{down}(x_i)V(t{+}1,x_{i-1}).$$

We define the normalized value function

$$v(t,i) := \frac{V(t,x_i)}{Q_0} - P_0.$$

Then the DPP becomes

$$v(t,i) = \max\Bigg[-\frac{s}{2}, -\frac{3s}{2}p(x_i) + \frac{s}{2}\pi(x_i){+}$$

$$\big(1{-}p(x_i){-}\pi(x_i)\big)\Big(q_{up}(x_i)v(t{+}1,i{+}1){+}(1{-}q_{up}(x_i){-}q_{down}(x_i))v(t{+}1,i){+}q_{down}(x_i)v(t{+}1,i{-}1)\Big)\Bigg],$$

with terminal condition

$$v(T,i) = -\frac{s}{2}, \qquad i = 1, \ldots, k.$$

The optimal stopping rule in feedback form, denoted $\nu^*(t,x)$, is characterized as follows: if execution has not yet occurred, stop at time t if and only if the maximum above is attained at the immediate execution value $-s/2$. Equivalently, one stops whenever

$$v(t,i) = -\frac{s}{2}, \qquad x_i = I_t.$$

In particular, at time $t = T-1$ we have

$$v(T-1,i) = \max\left[-\frac{s}{2}, -\frac{3s}{2}p(x_i) + \frac{s}{2}\pi(x_i) + (1 - p(x_i) - \pi(x_i))\left(-\frac{s}{2}\right)\right],$$

which simplifies to

$$v(T-1,i) = \max\left[-\frac{s}{2}, -\frac{s}{2} + s(\pi(x_i) - p(x_i))\right].$$

Therefore, if $\pi(I_{T-1}) \leq p(I_{T-1})$, it is optimal to stop at time $T-1$; otherwise, one waits until time T. Since $\pi(x)$ is increasing in x and $p(x)$ is decreasing in x, there exists a threshold x^* such that stopping at time $T-1$ is optimal if and only if $I_{T-1} \leq x^*$.

More generally, one can show that for every $t = T-1, \ldots, 0$, the function $v(t, \cdot)$ is increasing. Consequently, for each time t there exists a threshold $x^*(t)$ such that it is optimal to stop whenever $I_t \leq x^*(t)$.

The set

$$\mathcal{S} = \{(t, x) : x \leq x*(t),\ t = 0, \ldots, T-1\}$$

is called the stopping region, while its complement is referred to as the continuation region.

2.4 Model 1 Results

We ran our model via backtesting on several days of data on Microsoft (MSFT) and Intel (INTC) stock. The model was compared against two trivial benchmark strategies: one where an MO is submitted immediately, crossing the spread, and another where the LO is held until the terminal time unless otherwise filled, with the spread only crossed if the LO remains unfilled by the deadline.

Each strategy was evaluated relative to the midprice. The Always MO strategy incurs an immediate cost of half the spread (-\$0.005 in our setting), since the order executes at the best bid, which is by definition half a spread below the midprice. In contrast, the Always LO strategy has the opportunity to gain half a spread by being filled at the best ask, but faces the risk of adverse selection. If the price moves downward before the order is filled, the trader must cross the spread at a worse price, resulting in a loss of roughly one and a half spreads relative to the original midprice.

The stopping policy we previously discussed attempts to balance these competing objectives by using limit order imbalance as a predictor of short-term price movements. When the imbalance suggests that the probability of an unfavorable price movement is sufficiently high, the policy cancels the outstanding limit order and submits a market order instead. Intuitively, such a strategy should outperform both naive alternatives by avoiding unnecessary spread-crossing while reducing exposure to adverse price movements.

The empirical results only partially support this expectation. For each dataset, we trained the model on one day worth of data and then tested it on the next. Across all datasets tested, the policy consistently and comfortably outperformed the Always MO benchmark, confirming that waiting for favorable market conditions before crossing the spread has measurable value. However, as shown in Figures 6 and 7, the policy slightly underperformed the Always LO strategy over long horizons.

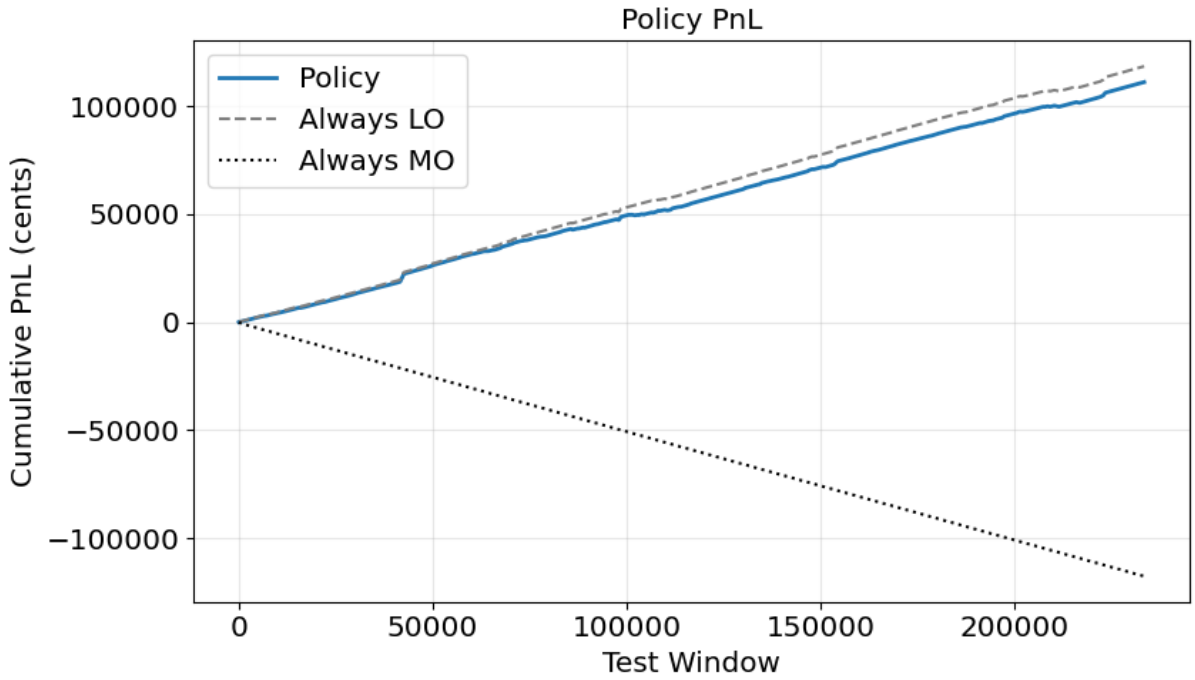


Figure 6: Cumulative PnL over many testing windows - trained on MSFT stock data from November 3, 2014 and tested on MSFT stock data from November 4, 2014.

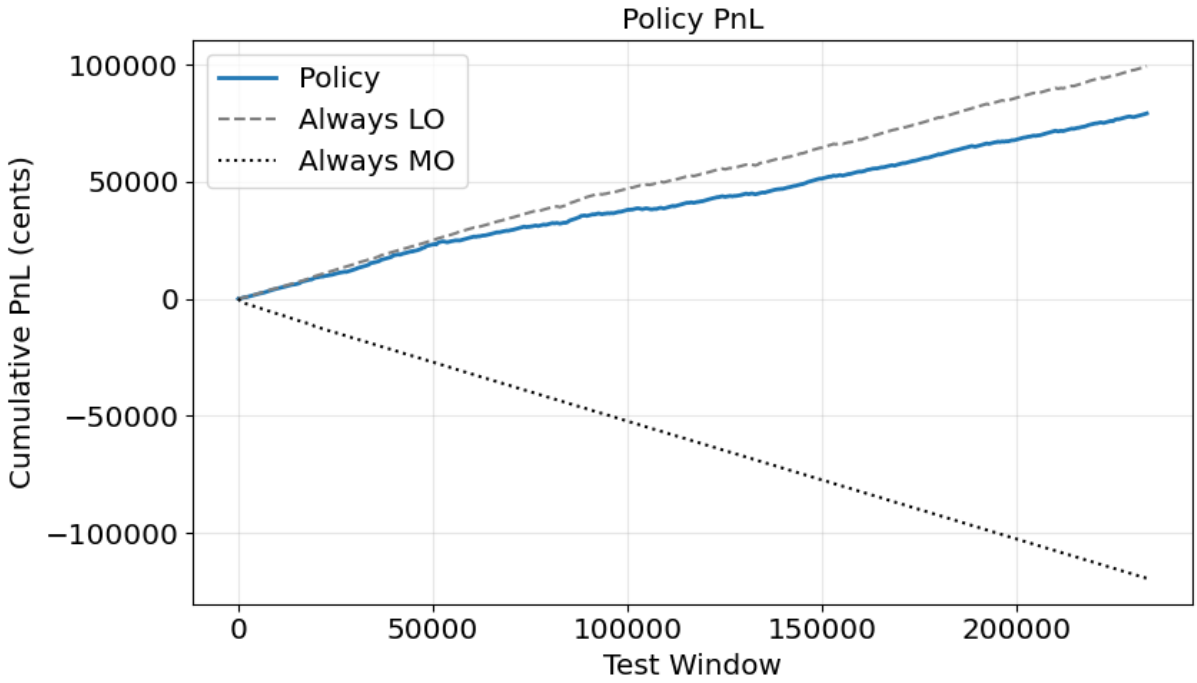


Figure 7: Cumulative PnL over many testing windows - trained on INTC stock data from November 13, 2014 and tested on INTC stock data from November 14, 2014.

At first glance, this result appears surprising. We would expect a strategy that reacts to order book information to dominate a strategy that simply waits passively. Upon closer examination, however, the discrepancy can largely be attributed to one of the simplifying assumptions in our

simulation. We assume that whenever a market order reaches our quoted price, our limit order is filled immediately, effectively placing our order at the front of the queue. In reality, limit orders are executed according to price-time priority, and an order submitted at the best price is often behind a substantial amount of resting volume. Consequently, many market orders would only partially consume the queue ahead of us, leaving our order unfilled.

This optimistic queue-position assumption substantially benefits the Always LO benchmark, as it greatly overestimates its fill probability while ignoring the execution delays that occur in real markets. Because the benchmark receives unrealistically frequent passive fills, it captures much of the spread without incurring the corresponding inventory or queue risk. Under these assumptions, the incremental advantage of switching from a limit order to a market order becomes much smaller, explaining why our policy is unable to consistently outperform the Always LO strategy.

Despite this limitation, the results remain encouraging. The policy consistently reduces the execution cost relative to immediately crossing the spread, demonstrating that order book imbalance contains meaningful predictive information about short-term market dynamics.

For the infinite-horizon case, the dynamic programming propagator is a contraction; the proof is given in Appendix 4.2. This implies that repeated application of the propagator converges to a time-independent fixed point v^{∞}. Therefore, in the $T = \infty$ model, the stopping decision does not depend on the remaining time horizon. It depends only on the current imbalance state I_t.

In the backtesting, the finite-T models use the time-dependent policy obtained by backward recursion. By contrast, the $T = \infty$ model uses the stationary policy obtained from the fixed point v^{∞}.

2.5 More Accurate Transition Models

We ran Monte Carlo simulations using our transition probabilities q_{up} and q_{down} to analyze how effectively they reflected the true values, starting at an arbitrary state and attempting to reproduce the dataset in terms of imbalance levels.

The overall goal was to determine if the simple transition probabilities were sufficient in modeling the dynamics of the process. We started the process at a random initial state and simulated 234,000 transitions, matching the total number of data points in our sample. Figure 5 shows the results of this simple transition model. The states near the middle are represented relatively well, but there appears to be underrepresentation near either extreme. This suggests that we should change our approach in modeling transitions.

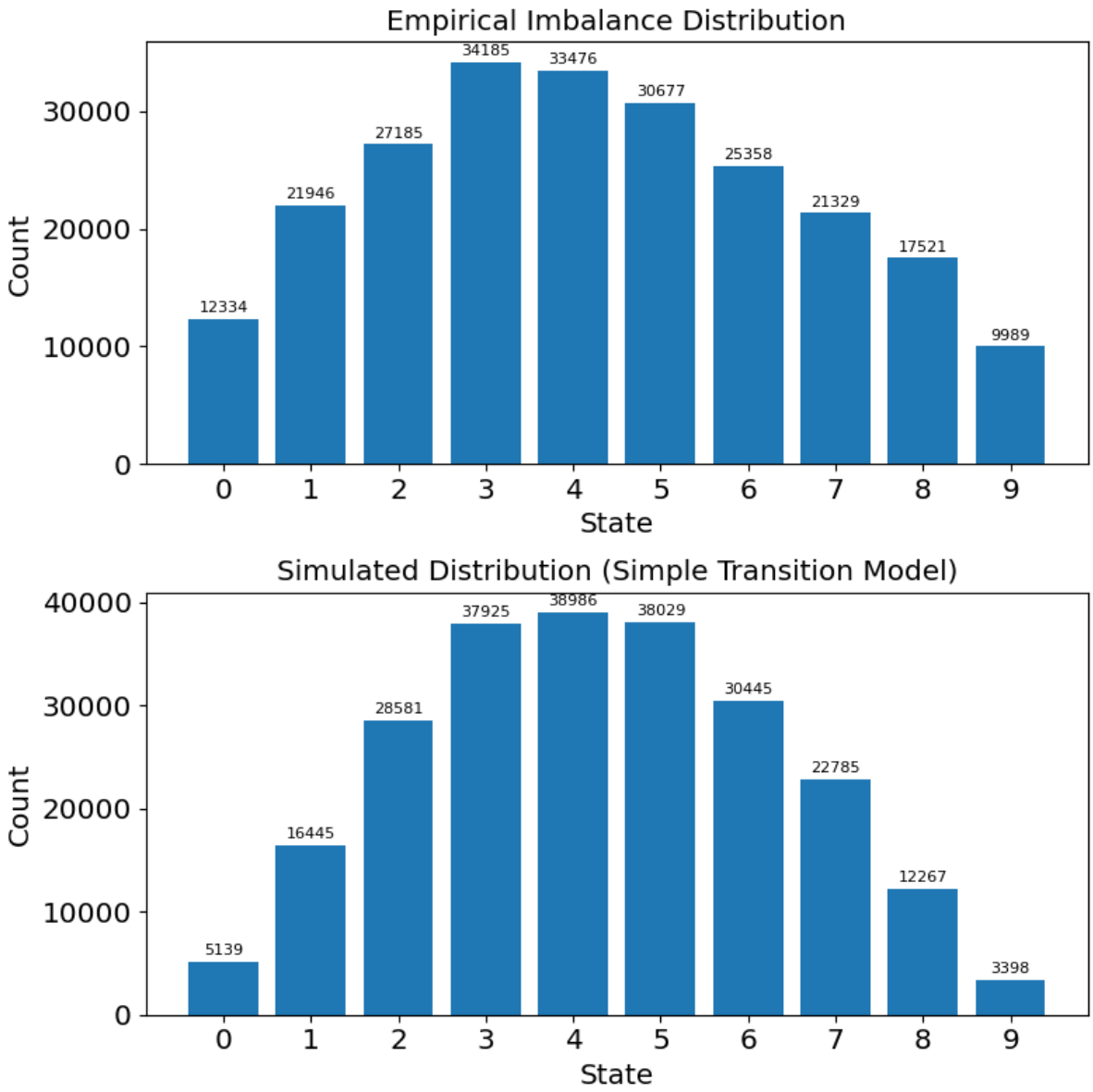


Figure 8: Monte Carlo simulation for our simple model, using empirically estimated q_{up} and q_{down} (Microsoft data from November 21, 2014).

We decided to use a full transition matrix, obtaining a probability $q(i, j)$ for each transition from $I = x_i$ to $I = x_j$. Since the required sample sizes for probability estimation are only dependent on state frequency, we are free to estimate these probabilities using the same samples.

Running the same Monte Carlo simulation for 234,000 transitions using the full transition matrix produces results seen in Figure 6. This appears to reflect the actual imbalance dynamics much more accurately than the simplistic model.

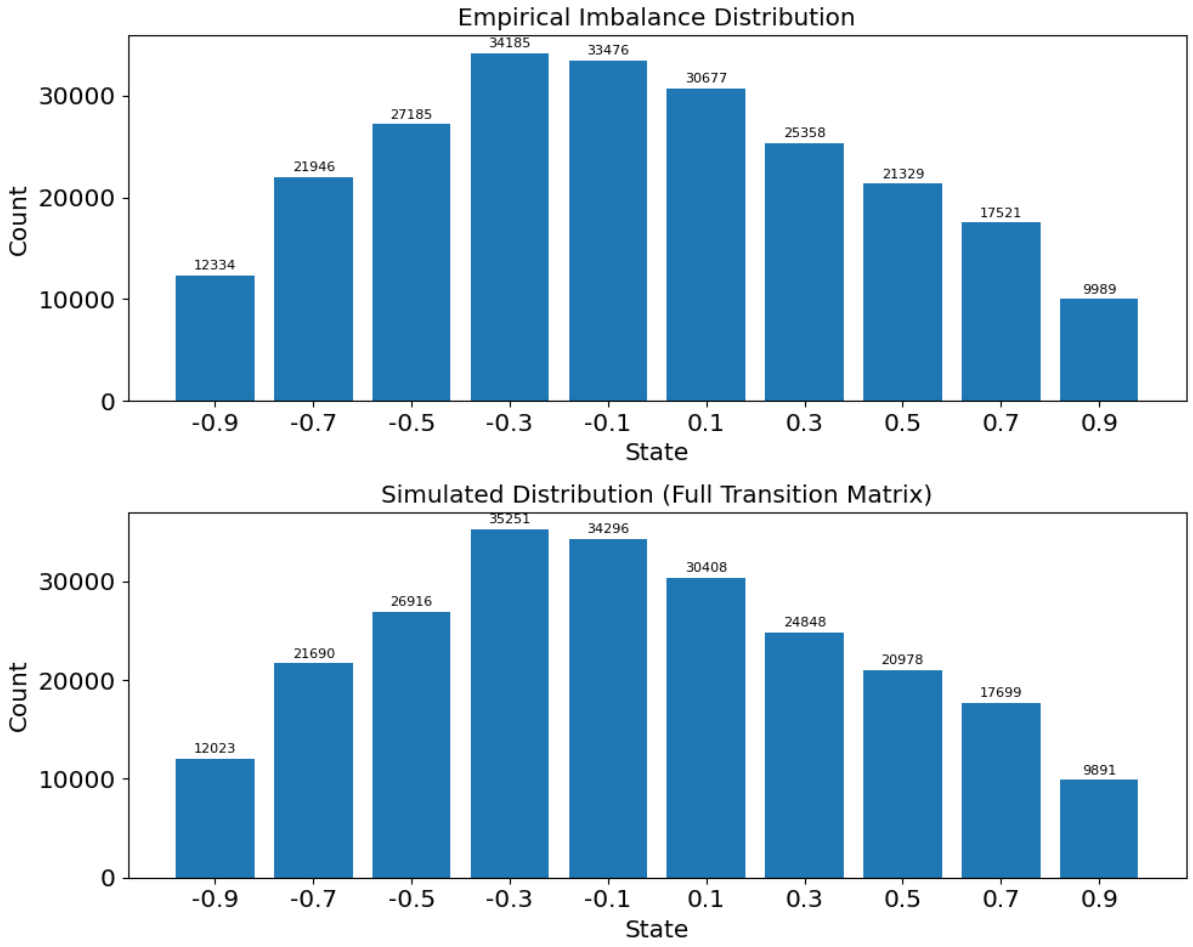


Figure 9: Monte Carlo simulation for our full probability model, using an empirically estimated transition matrix (Microsoft data from November 21, 2014).

2.6 Initial Model Results with Full Transition Matrix

Updating the backtesting to use our more realistic full-matrix transition model, we ran our tests again to determine whether the richer transition dynamics would meaningfully affect the optimal stopping policy. Unlike the original approximation, which only permits transitions between neighboring imbalance states, the full transition matrix captures the empirical probability of moving between any pair of imbalance states. Although large jumps are relatively uncommon, they nevertheless contribute to the expected continuation value used in the dynamic programming recursion.

As shown in Figure 10, incorporating these additional transitions noticeably changes the computed stopping boundary. The policy becomes more responsive to regions of the state space where imbalance is likely to change rapidly, producing a stopping rule that better reflects the observed behavior of the order book. Rather than relying on a purely local approximation of the imbalance process, the full transition model accounts for the complete distribution of future imbalance states when evaluating the trade-off between continuing to wait for a passive fill and immediately crossing the spread.

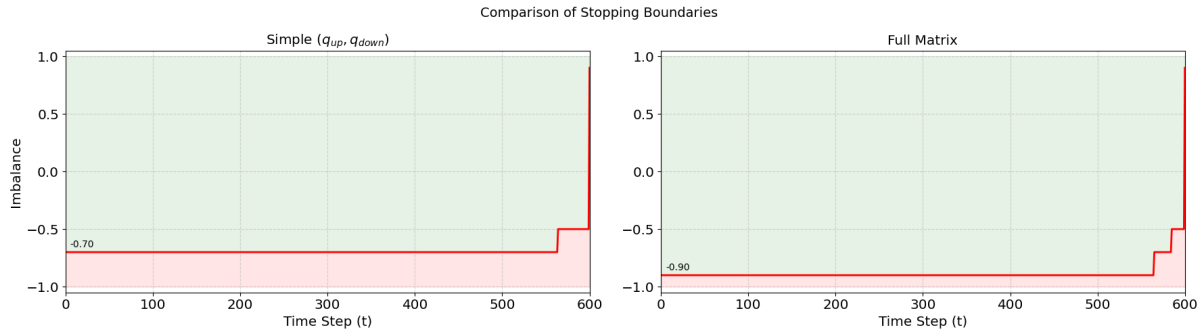


Figure 10: Stopping boundaries for simple and full transition models, MSFT stock, November 7, 2014.

These changes are also reflected in the backtesting results shown in Figures 11 and 12. While the improvement in cumulative PnL is modest, the full transition model consistently outperforms the simplified q_{up}, q_{down} approximation. This suggests that even relatively infrequent long-range transitions contain useful information when computing the continuation value. By more accurately modeling the evolution of the limit order book, the policy makes better cancellation decisions and avoids some unnecessary market orders that arise under the simplified model.

The improvement is not dramatic because most imbalance transitions are still between neighboring states, meaning that the simple transition approximation captures much of the underlying dynamics. Nevertheless, the full transition matrix consistently produces stopping boundaries that are more closely aligned with the empirical behavior of the market and therefore yields a small but measurable improvement in execution performance.

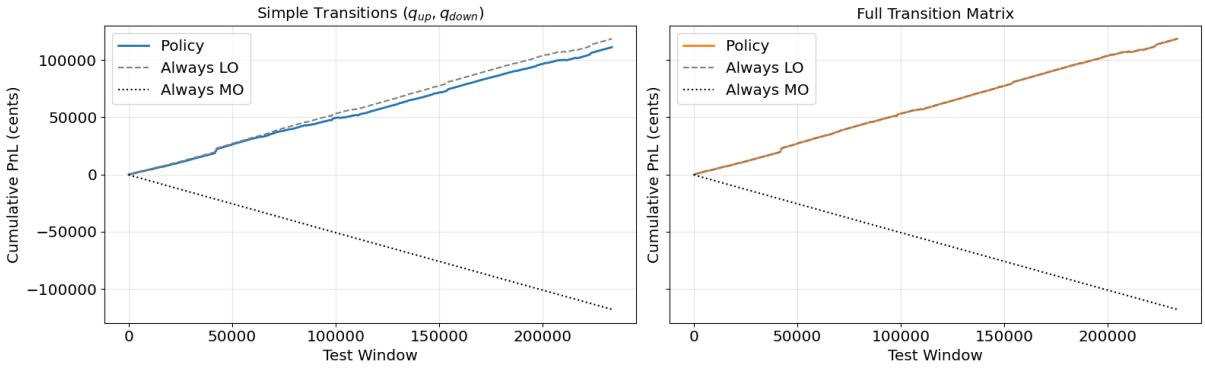


Figure 11: Cumulative PnL over many testing windows using two different transition models - trained on MSFT stock data from November 3, 2014 and tested on MSFT stock data from November 4, 2014.

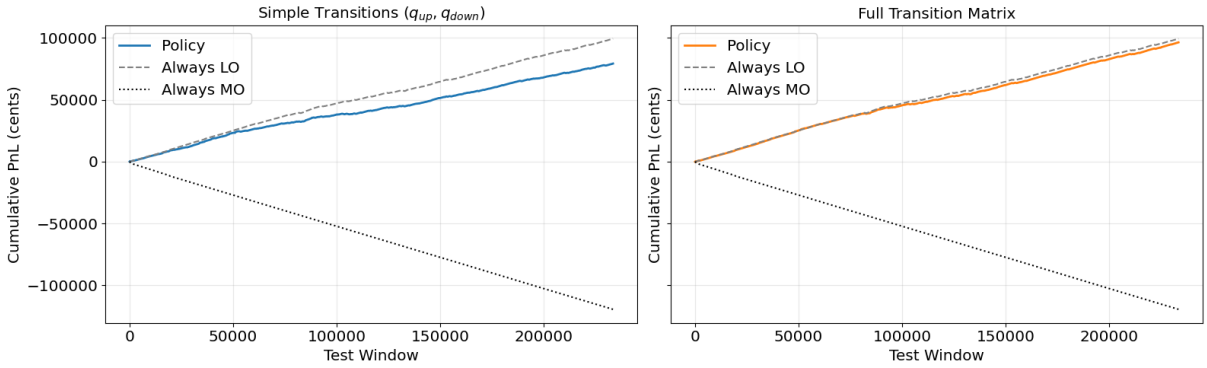


Figure 12: Cumulative PnL over many testing windows using two different transition models - trained on INTC stock data from November 13, 2014 and tested on INTC stock data from November 14, 2014.

3 A Refined Model

3.1 Queue Position as an Additional State Variable

The main weakness of the initial model is that it implicitly assumes that our posted sell limit order has immediate priority at the best ask. In reality, when we submit a sell LO at the current best ask, there is already some existing sell volume ahead of us. Therefore, our order will only be executed after the queue ahead of us has been depleted by incoming buy market orders or by cancellations of sell limit orders ahead of us.

To account for this, we extend the state process from only imbalance to a two-dimensional process

$$Z_t = (I_t, M_t),$$

where I_t is the discretized imbalance state and M_t is the discretized queue position state. Here M_t represents the amount of sell-side volume ahead of our order at the best ask. If M_t is large, our limit order is far back in the queue and is unlikely to be filled quickly. If M_t is close to zero, our order is near the front of the queue.

We discretize the queue position into equally spaced states

$$M \in \{0, \Delta M, 2\Delta M, \ldots, M_{\max}\}.$$

In our implementation, we take

$$\Delta M = 100,$$

so that one queue state corresponds to one order of 100 shares. This is also the order size used in the backtest. The full state space is therefore

$$\mathcal{Z} = \mathcal{X} \times \mathcal{M},$$

where $\mathcal{X}$ is the imbalance grid and $\mathcal{M}$ is the queue-position grid.

3.2 Queue Movement and Market Orders

The queue ahead of our sell limit order decreases when buy market orders arrive at the best ask. If a buy market order of volume BMO_t arrives during the transition from time t to time $t+1$, then the queue ahead of us decreases by approximately that amount. If the buy market order is large enough to consume all volume ahead of us and our own order, then our LO is filled.

If our order size is Q_0, then the fill condition is

$$BMO_t \geq M_t + Q_0.$$

If this condition is not met, then the updated queue position is approximately

$$M_{t+1} = M_t - BMO_t.$$

Since the model is discretized, this continuous queue movement is mapped back onto the nearest queue state in $\mathcal{M}$.

3.3 Cancellation Proxy

In addition to buy market orders, cancellations can also move us closer to the front of the queue. If sell limit orders ahead of us are canceled, then the number of shares ahead of our order decreases. However, cancellations do not execute our order; they only improve our queue position.

We do not directly observe which cancellations occur ahead of our hypothetical order. Therefore, we construct a cancellation proxy using changes in best ask volume. When the best ask price is unchanged, we decompose the change in best ask volume as

$$V^a_{t+1} = V^a_t - BMO_t - C_t + LO^{new}_t,$$

where V^a_t is the best ask volume, C_t is cancellation volume, and LO^{new}_t represents newly submitted sell limit orders at the best ask.

Rearranging gives

$$V^a_t - V^a_{t+1} - BMO_t = C_t - LO^{new}_t.$$

Since LO^{new}_t is not separately observed, we define the net cancellation proxy

$$C^{proxy}_t = \max\left(0, V^a_t - V^a_{t+1} - BMO_t\right),$$

only when the best ask price remains unchanged. This quantity is not the true cancellation volume. It should be interpreted as a conservative proxy, because new sell LOs can hide cancellations.

3.4 Relative Queue Position

A fixed queue position can mean very different things depending on the total size of the best ask queue. Being 3000 shares behind is very different when the total best ask volume is 4000 shares versus 50000 shares. Therefore, instead of assuming that the cancellation effect depends only on the absolute queue state M_t, we use the relative queue position

$$r_t = \frac{M_t}{V^a_t}.$$

We model the fraction of inferred cancellations that occur ahead of us by a function

$$g(r_t) = r_t^{\gamma},$$

where $\gamma > 1$ means that cancellations are more likely to benefit us when we are farther back in the queue. The cancellation-adjusted queue movement is therefore

$$Y_t = BMO_t + \kappa C^{proxy}_t g(r_t),$$

where κ controls how strongly the cancellation proxy is incorporated. If $\kappa = 0$, then cancellations are ignored and only buy market orders move us forward.

Thus, the queue position update is

$$M_{t+1} = \max(M_t - Y_t, 0),$$

unless the buy market order is large enough to fill our own order.

Partial observability. The use of the relative queue position introduces an additional state variable into the queue dynamics. In particular, the queue update depends on

$$r_t = \frac{M_t}{V_t^a},$$

and therefore depends not only on the observed state (I_t, M_t), but also on the best ask queue size V_t^a. Thus, the process (I_t, M_t) is not necessarily Markov by itself. The Markov state would instead be the enlarged process

$$\widetilde{Z}_t = (I_t, M_t, V_t^a),$$

or equivalently (I_t, M_t, r_t).

In the implementation, however, we keep the lower-dimensional state

$$Z_t = (I_t, M_t)$$

for tractability. The variable V_t^a is therefore treated as a hidden or partially observed component of the state. The transition probabilities estimated conditional on (I_t, M_t) should be interpreted as projected, or averaged, transition probabilities:

$$\mathbb{P}(Z_{t+1} \mid Z_t) = \sum_v \mathbb{P}(Z_{t+1} \mid Z_t, V_t^a = v)\mathbb{P}(V_t^a = v \mid Z_t).$$

Thus, the frequency estimates used in the finite-state dynamic program average over the empirical distribution of best ask queue sizes observed at each reduced state (I_t, M_t). This is an approximation to the fully Markov model on the enlarged state space. When $\kappa = 0$, the cancellation term is removed, and this particular source of partial observability disappears.

3.5 Transition Probabilities in the Secondary Model

The imbalance transition probabilities $q_{up}(x)$ and $q_{down}(x)$ are estimated in the same way as in the initial model. The new component is the queue transition probability. For each state (x, m), we estimate

$$\lambda(x, m, y) = \mathbb{P}(\text{queue moves forward by } y \text{ states} \mid I_t = x, M_t = m).$$

The queue movement y is computed from the effective queue movement Y_t by rounding to the nearest queue grid size:

$$y_t = \text{round}\left(\frac{Y_t}{\Delta M}\right).$$

The empirical distribution of these movements gives $\lambda(x, m, y)$.

The fill probability is then obtained from the probability that the queue movement is larger than the current queue position:

$$\pi(x, m) = \sum_{y > m} \lambda(x, m, y).$$

This represents the probability that the queue ahead of our order is fully depleted and our LO is filled in the next time step.

3.6 Dynamic Programming Principle for the Secondary Model

The normalized value function is now defined over the joint state:

$$v(t, x, m).$$

The immediate stopping payoff is still

$$-\frac{s}{2},$$

because stopping means canceling the LO and submitting a sell MO at the best bid.

The continuation value has three components:

- the adverse price-move event, with payoff $-\frac{3s}{2}$,
- the LO fill event, with payoff $+\frac{s}{2}$,
- the non-terminal event, where the process transitions to a new state (x', m').

Thus, the secondary model DPP is

$$\begin{aligned} v(t,x,m) = \max\Bigg\{ & -\frac{s}{2}, \\ & -\frac{3s}{2}p(x) + \frac{s}{2}\pi(x,m) \\ & + (1 - p(x) - \pi(x,m))\mathbb{E}\Big[v(t+1, I_{t+1}, M_{t+1}) \mid I_t = x, M_t = m\Big]\Bigg\}. \end{aligned}$$

Note that $p(x)$ does not depend on m since the best bid dropping does not depend on where we are in the best ask queue.

Using the estimated imbalance and queue transition probabilities, the expectation is computed as

$$\mathbb{E}\left[v(t+1, I_{t+1}, M_{t+1})\right] = \sum_{x'} \sum_{m'} \mathbb{P}(I_{t+1} = x' \mid I_t = x)\mathbb{P}(M_{t+1} = m' \mid I_t = x, M_t = m) v(t{+}1, x', m').$$

The terminal condition remains

$$v(T, x, m) = -\frac{s}{2}$$

for all states (x, m), since at the terminal time we submit a market order if the LO has not already been filled.

3.7 Backtesting the Secondary Model

In the backtest, the DPP policy uses the discretized state (x, m), but the simulated queue position is tracked in shares rather than only as a discrete state. At the beginning of each simulated trade, we submit a sell LO at the initial best ask. Since all currently posted ask volume is ahead of us, the initial queue position is

$$M_0^{real} = V_0^a.$$

At each time step, the continuous queue position is updated using the observed buy MO volume and the cancellation proxy. The continuous queue position is then mapped to the nearest discrete queue state m only when querying the DPP policy.

The LO is filled only if aggressive buy volume clears the queue ahead of us and our order size:

$$BMO_t \geq M_t^{real} + Q_0.$$

Cancellations can reduce M_t^{real}, but cancellations alone do not execute our order.

If the historical best ask level disappears upward without our hypothetical order being filled, the counterfactual book differs from the observed historical book. Since the historical LOB does not include our order, the disappearance of the old best ask does not necessarily imply that our hypothetical LO would also disappear. In the counterfactual book, our order remains posted at the old ask price and becomes first in the queue. We therefore set

$$M_t^{real} = 0.$$

Once our order is first in the queue, it is executed as soon as either a buy market order arrives or a buy limit order is posted at our limit price. Since our order size is set to the minimum order size, any such incoming buy market order is large enough to execute our order. Similarly, a buy limit order submitted at the price level occupied by our sell limit order is treated as marketable against our order and therefore results in execution.

The secondary model is compared against two benchmark strategies:

- Always MO: immediately submit a market order at the current best bid.
- Always LO: submit a limit order at the current best ask and wait, subject to the same queue-position and adverse-price-move logic.

3.8 Backtesting Results for the Secondary Model

We implemented the secondary model on MSFT using $T = 600$, one-day train/test splits, and $\kappa = 0$. The policy was evaluated against the Always-MO and Always-LO benchmarks using the queue-aware backtesting procedure described above.

Figure 13 shows the optimal stopping boundaries obtained from the dynamic program. Each subplot fixes a queue-ahead state M_t and plots the imbalance threshold separating the stopping and continuation regions. The policy becomes more aggressive as the queue position increases: when more volume is ahead of our order, passive execution is less likely, so the model is willing to stop under less adverse imbalance conditions. When the order is closer to the front of the queue, the continuation region is larger because waiting has a higher probability of producing a passive fill.



Figure 13: Optimal stopping boundaries for the secondary model. Each panel fixes a queue-ahead state and shows the stopping threshold as a function of time.

The cumulative PnL curves in Figure 14 show that the learned policy improves on both naive benchmarks. The Always-MO strategy performs poorly because it always crosses the spread immediately. The Always-LO strategy avoids this immediate cost, but it is exposed to adverse price movements while waiting in the queue. The secondary-model policy balances these two effects by waiting when passive execution is attractive and switching to a market order when the estimated continuation value becomes unfavorable.

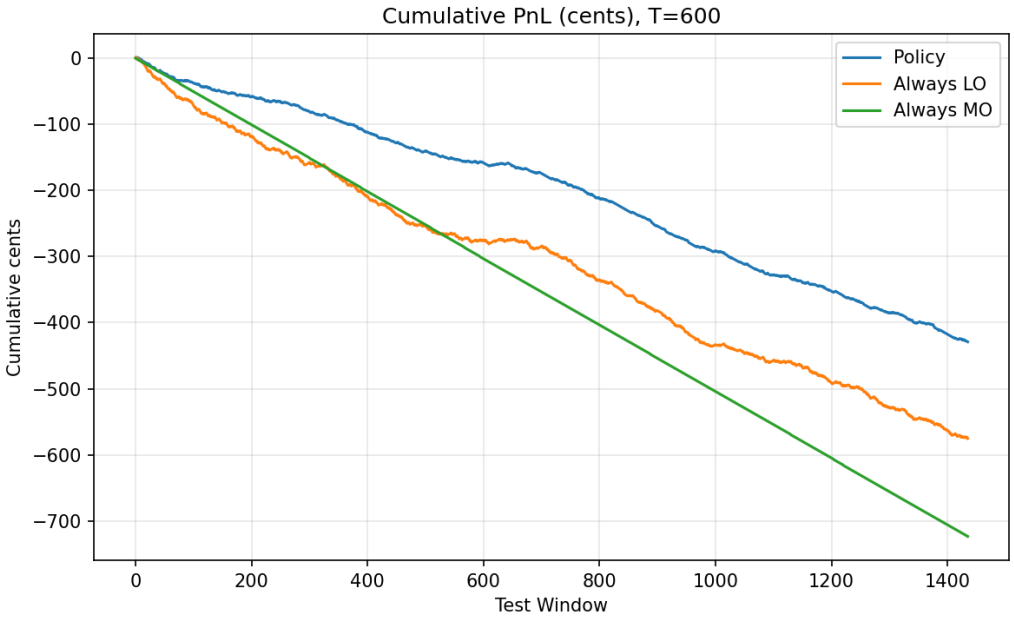


Figure 14: Cumulative PnL of the secondary-model policy compared with the Always-MO and Always-LO benchmarks.

This behavior is also visible in the PnL distribution shown in Figure 15. The outcomes are concentrated around a few values because most executions correspond to crossing the spread, receiving a passive fill, or exiting after an adverse price movement. The policy distribution is shifted closer to zero than the two benchmark distributions, indicating a lower average implementation shortfall.

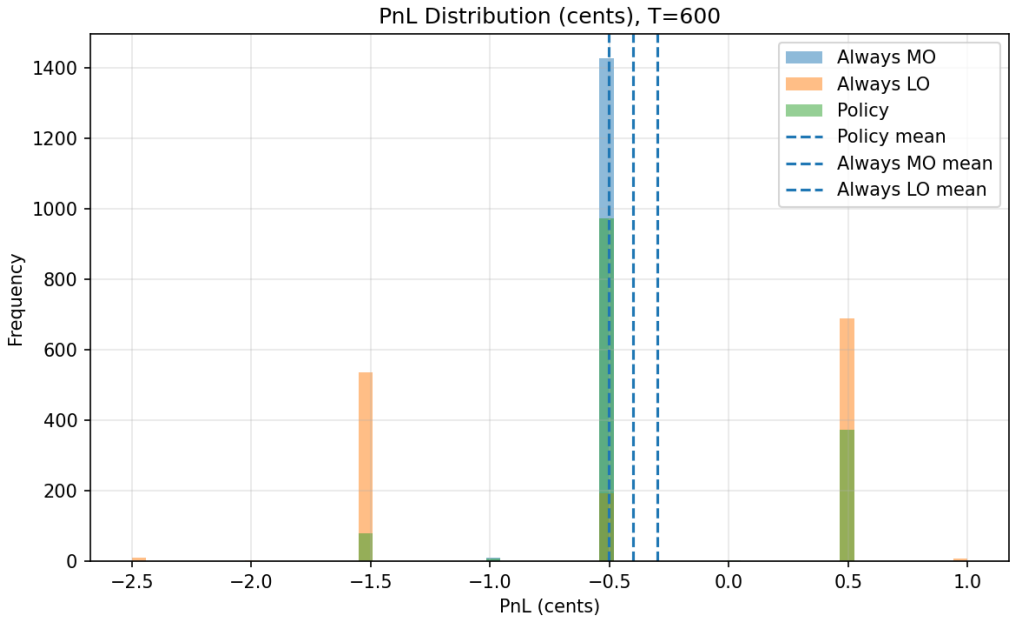


Figure 15: Distribution of per-trade PnL for the secondary-model policy and benchmark strategies. Dashed lines indicate sample means.

Quantitatively, the policy achieves a mean PnL of −0.299 cents per share, compared with −0.503 cents for Always MO and −0.400 cents for Always LO. The policy stops by submitting a market order in 66.7% of test windows and obtains passive limit-order fills in 26.0% of test windows. These results suggest that incorporating queue position into the state substantially improves the realism of the execution model and allows the stopping policy to outperform both simple benchmarks.

3.9 Experiments

3.9.1 Sensitivity to the Execution Deadline T

To isolate the effect of the execution deadline itself, the DP policy was re-solved and backtested for nine deadline values, $T \in \{50, 100, 250, 500, 600, 1000, 2000, 5000, \infty\}$ seconds, holding the transition model, state space, and risk-aversion parameter fixed. Each T was trained and backtested on the same four train/test day pairs (`20141103/04`, `20141105/06`, `20141110/11`, `20141112/13`), pooled across pairs, for a total of 35,392 trades across all nine configurations.

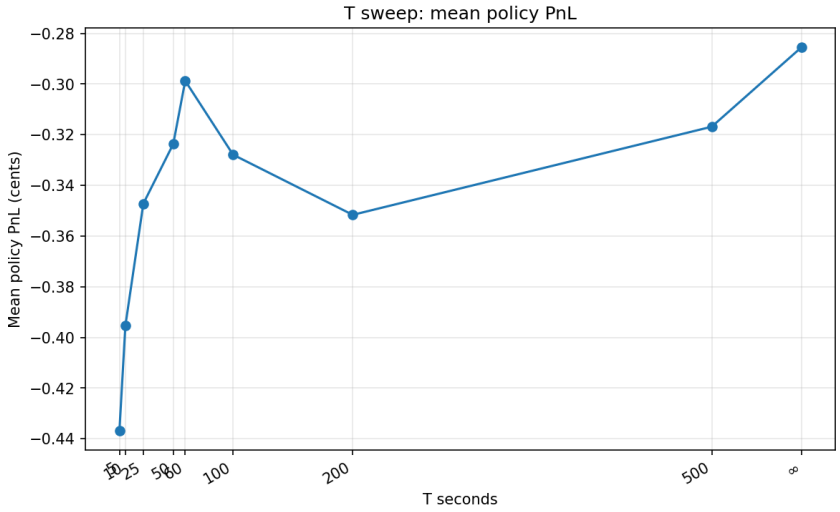


Figure 16: Mean policy PnL as a function of the execution deadline T.

Table 1: Backtest results by deadline T (seconds; ∞ = no deadline).

T (s)	n (windows)	Policy (¢)	Stop rate	LO fill rate	Deadline rate
50	17276	−0.437	0.721	0.076	0.194
100	8636	−0.395	0.723	0.125	0.135
250	3452	−0.347	0.711	0.195	0.056
500	1724	−0.324	0.669	0.245	0.021
600	1436	−0.299	0.667	0.260	0.017
1000	860	−0.328	0.673	0.249	0.007
2000	428	−0.352	0.689	0.236	0.000
5000	172	−0.317	0.640	0.273	0.000
∞	1408	−0.286	0.650	0.283	0.000

Mean policy PnL is non-monotonic in T (Figure 16). It improves sharply as the deadline is relaxed from $T = 50$ s (−0.437¢) to $T = 600$ s (−0.299¢, the best finite-T result), worsens again

through $T = 1000$–$2000\,\text{s}$ (bottoming at $-0.352¢$), and then improves a second time out to $T = 5000\,\text{s}$ and $T = \infty$, which gives the best result overall at $-0.286¢$.

The initial improvement (short T up to $T = 600$) tracks a clear behavioral mechanism: the deadline rate – trades forced to execute because the horizon expired before a stop or fill condition triggered – falls from 19.4% at $T = 50\,\text{s}$ to 1.7% by $T = 600\,\text{s}$ and to exactly 0% for $T \geq 2000\,\text{s}$, while the limit-order fill rate rises correspondingly from 7.6% to a peak of 28.3% at $T = \infty$. At short deadlines the policy is frequently cut off before its resting limit order can fill and is forced into an unfavorable execution; giving it more time converts many of these forced exits into either passive fills or better-timed market orders.

The mid-range dip at $T = 1000$–$2000\,\text{s}$ is harder to attribute to this same mechanism, since the deadline rate is already at or near zero by that point. It coincides with a substantial drop in sample size ($n = 860$ and $n = 428$, versus $n = 17{,}276$ at $T = 50\,\text{s}$ and $n = 1408$ at $T = \infty$), since fewer non-overlapping T-second windows fit inside the fixed four-day test sample as T grows. Given how few windows back the $T = 1000$–$5000\,\text{s}$ estimates, this dip should be treated as a candidate sampling artifact rather than firm evidence of a genuine non-monotonicity in the underlying policy value, and would benefit from a larger backtest sample before drawing a stronger conclusion.

3.9.2 Cross Ticker Evaluation

The second model was evaluated across multiple equities to assess its robustness beyond MSFT. Since the dynamic programming formulation assumes a constant one-tick spread and single-tick price movements, we first verified these assumptions on all ten LOBSTER tickers over eighteen trading days in November 2014.

The stocks naturally separated into three groups. Six tickers (MSFT, INTC, MU, VOD, FB, and EBAY) satisfy the assumptions well, with one-tick spreads occurring approximately 92–99% of the time and one-tick price changes approximately 90–98% of the time. AMZN, GOOG, and PCAR were excluded because their spreads frequently exceeded one tick (median spreads of 11 and 17 ticks for AMZN and GOOG, respectively, while PCAR exhibited a one-tick spread only about 52% of the time). SMH was considered borderline, with a one-tick spread only about 65% of the time. Consequently, the experiment was conducted using only the top four suitable tickers.

Figure 17 compares the mean implementation shortfall for four representative stocks. The proposed policy achieves mean shortfalls of approximately -0.272, -0.312, -0.330, and -0.407 cents for MSFT, INTC, MU, and VOD, respectively.

Although execution costs differ across securities, the policy performs consistently on all stocks that satisfy the model assumptions. More liquid stocks, such as MSFT, provide more opportunities for passive execution, resulting in lower implementation shortfall, whereas less liquid stocks, such as VOD, require more aggressive execution and consequently incur higher costs.

3.9.3 Effect of the Cancellation Allocation Model

Our original queue-aware model includes a parameter κ that controls the influence of order cancellations on the effective queue ahead. Since the true locations of cancellations are unobservable in historical data, we approximate their impact by allocating cancellations according to the relative queue position

$$r_t(m) = \frac{M_t}{V_t^a},$$

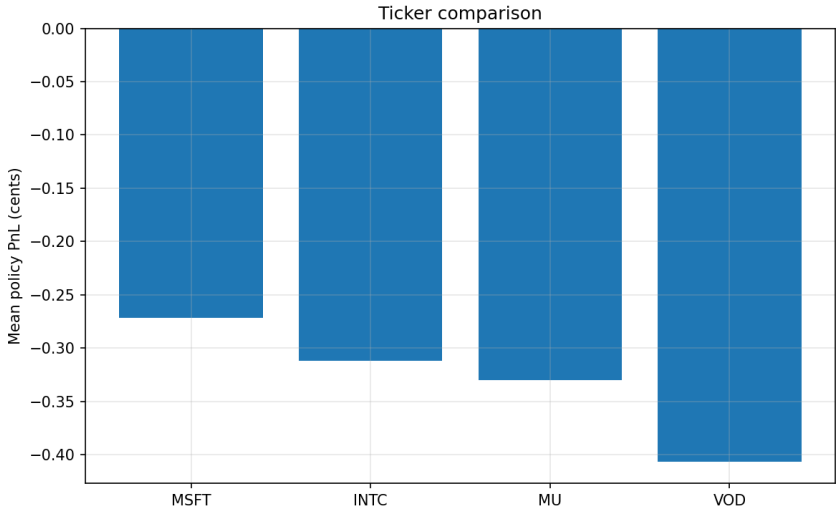


Figure 17: Mean implementation shortfall across representative tickers.

using the weighting function

$$g(r_t(m)) = r_t(m)^\gamma,$$

with $\gamma = 2$. When $\gamma > 1$, cancellations are concentrated toward the back of the queue, so cancellations near the front have relatively little effect while those occurring deeper in the queue contribute more to queue advancement.

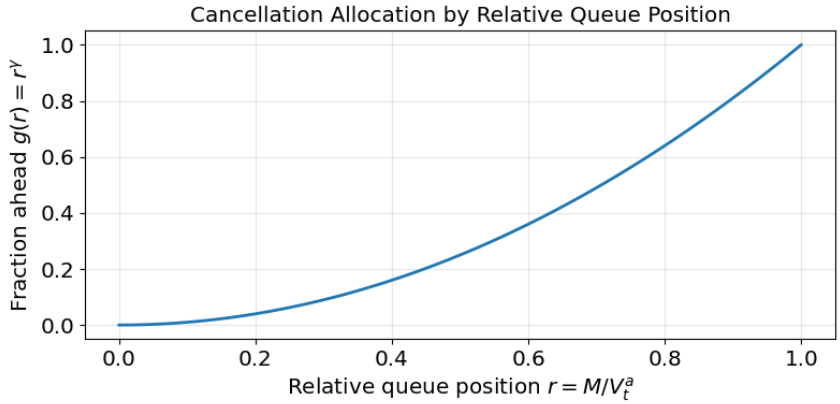
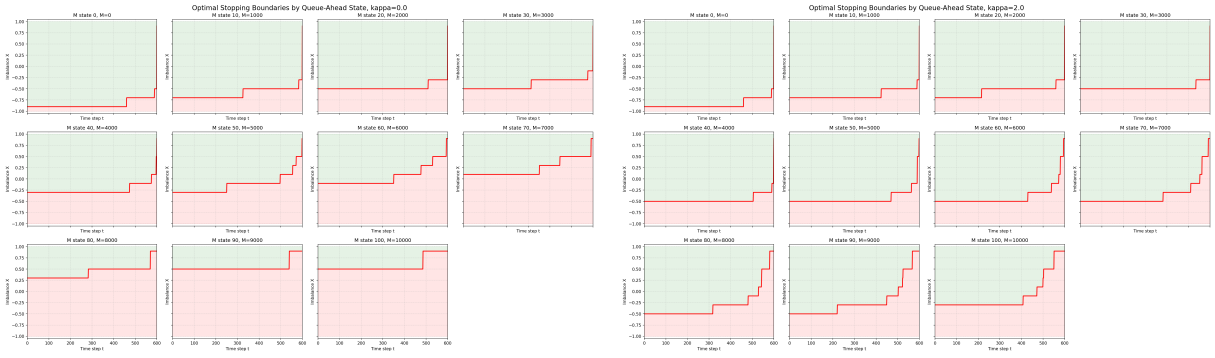


Figure 18: Cancellation allocation function $g(r) = r^2$ used to model the fraction of cancellations occurring ahead of the order as a function of relative queue position.

We compare two settings. The baseline, $\kappa = 0$, assumes that all cancellations occur behind the order and therefore provide no benefit, representing a worst-case scenario. The second setting, $\kappa = 2$, incorporates the cancellation allocation model above.

Figures 19a and 19b compare the resulting optimal stopping boundaries. Incorporating cancellations lowers the stopping boundary across nearly all queue-ahead states, particularly for larger queue positions. Intuitively, when cancellations are expected to advance the queue, the probability of obtaining a passive fill increases, making it optimal to continue waiting under a wider range of imbalance states rather than immediately crossing the spread.

Figure 20 shows that incorporating the cancellation model ($\kappa = 2$) yields a modest improvement in mean implementation shortfall relative to the baseline $\kappa = 0$. This improvement is expected,



(a) Stopping boundary when $\kappa = 0$ **(b)** Stopping boundary when $\kappa = 2$

Figure 19: Optimal stopping boundaries under different cancellation models. The stopping boundary shifts downward when cancellations are incorporated ($\kappa = 2$), indicating that the policy is willing to wait longer for passive execution.

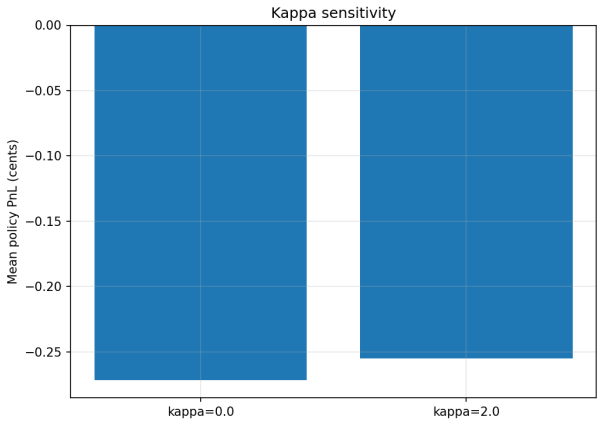


Figure 20: Mean PnL under the two cancellation models. Incorporating the heuristic cancellation allocation model ($\kappa = 2$) produces a modest improvement over the conservative baseline ($\kappa = 0$).

as accounting for cancellations increases the estimated probability of passive fills, allowing the policy to wait longer before crossing the spread.

However, we do not adopt the cancellation model in the final version of our framework. The allocation function $g(r) = r^\gamma$ is heuristic and introduces an artificial assumption about where cancellations occur within the queue, since historical data do not reveal their true positions. Moreover, incorporating κ modifies both the optimal policy and the backtesting simulator through the queue evolution model, making it difficult to attribute the observed performance improvement solely to a better execution strategy. To avoid introducing additional modeling assumptions, we therefore use the more conservative $\kappa = 0$ formulation as our primary model.

3.9.4 Train/Test Window Sensitivity

Six train/test window configurations were constructed from the 18-day MSFT sample (`20141103`–`20141126`), each with the first 30 minutes of every trading day excluded. They vary along two axes: window *length* and the *gap* between the training and test segments.

For $n \in \{1, 2, 3\}$, the model trains on the first n trading days (concatenated) and tests on the following n trading days: e.g. $n = 1$ trains on day 1 and tests on day 2; $n = 3$ trains on days 1–3 and tests on days 4–6. Train and test segments are always full trading days and are separated by

roughly one day.

For window lengths of 60, 180, and 720 minutes, the model trains on a window at the start of each trading day's usable data and tests on the immediately following window of the same length, with no gap between them. Results are pooled across all 18 days. The 60- and 180-minute windows fit within a single trading day. The 720-minute (12-hour) window does not fit twice inside one post-exclusion trading day, so that case instead concatenates all 18 days into one continuous stream and splits it into consecutive, non-overlapping 12-hour train/test blocks, some of which bridge across day boundaries.

Table 2: Summary of the six train/test window configurations.

Configuration	Train span	Test span	Train/test gap
`train_1d_test_1d`	1 day	next 1 day	∼1 day
`train_2d_test_2d`	2 days	next 2 days	∼1 day
`train_3d_test_3d`	3 days	next 3 days	∼1 day
`next_window_60min`	60 min (day start)	next 60 min, same day	none
`next_window_180min`	180 min (day start)	next 180 min, same day	none
`next_window_720min`	12h block (pooled)	next 12h block	none

Table 3: Mean PnL (cents) by train/test window configuration: policy vs. benchmarks.

Configuration	Policy	Always-MO	Always-LO
`train_1d_test_1d`	−0.272	−0.503	−0.519
`train_2d_test_2d`	−0.292	−0.503	−0.462
`train_3d_test_3d`	−0.328	−0.503	−0.404
`next_window_60min`	−0.349	−0.506	−0.458
`next_window_180min`	−0.352	−0.503	−0.422
`next_window_720min`	−0.345	−0.504	−0.481

Table 3 summarizes the mean implementation shortfall under six train/test window configurations. Across all settings, the proposed policy consistently outperforms both benchmark strategies. The mean implementation shortfall ranges from −0.272 cents for the 1-day rolling window to −0.352 cents for the 180-minute intraday window, compared with an almost constant −0.503 cents for the Always-MO benchmark and −0.404 to −0.519 cents for the Always-LO benchmark. These results indicate that the policy remains robust to the choice of training horizon.

Figure 21 illustrates the policy performance across the six configurations.

Although the policy consistently outperforms both benchmarks, performance degrades slightly as the training window lengthens and is weakest for the intraday configurations. This decline is accompanied by a change in execution behavior: the market-order stop rate increases from 60.7% under the 1-day configuration to 75.1% for the 180-minute configuration, while the passive limit-order fill rate decreases from 30.1% to 19.4%. In contrast, the fraction of executions triggered by adverse price movements remains relatively stable, varying between approximately 4% and 8% across all configurations. These results suggest that different training windows primarily influence how aggressively the policy chooses to execute, rather than substantially changing its exposure to adverse price movements. Because the sample sizes vary considerably across configurations ($n = 359$–3239), the overall trend is more reliable than the precise numerical differences.

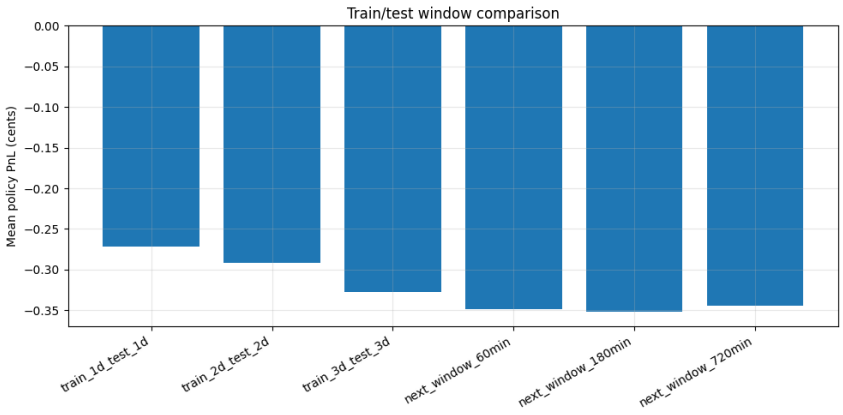


Figure 21: Mean implementation shortfall of the proposed policy under different train/test window configurations.

3.9.5 Stationary ($T = \infty$) Policy Evaluation

The finite-horizon DP policy imposes a fixed 60-second execution deadline ($T = 600$ steps); in this experiment, we tested whether removing this deadline changes policy behavior or performance by solving the stationary Bellman equation via value iteration and backtesting the resulting time-independent policy with no forced execution cutoff. The model was trained on `MSFT_20141103` and backtested on `MSFT_20141104` ($n = 360$ test windows).

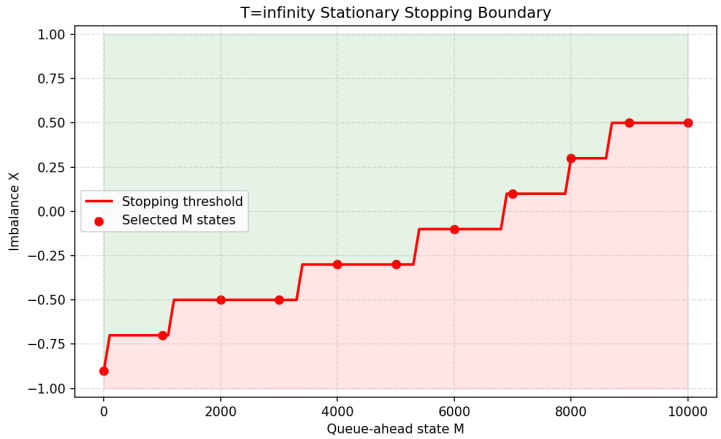
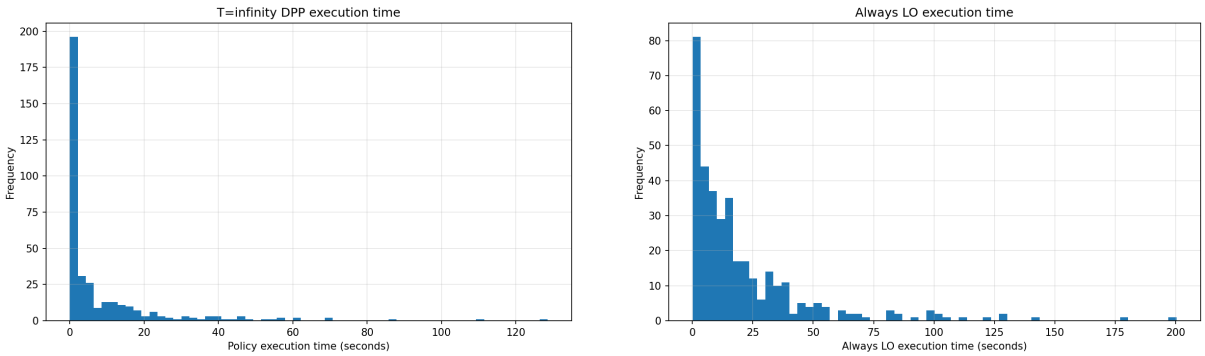


Figure 22: Stationary stopping boundary under $T = \infty$.

The stationary stopping boundary (Figure 22) is a monotonically increasing step function of queue-ahead state M: for orders near the front of the queue ($M \approx 0$), the policy only stops for severely adverse imbalance ($X \lesssim -0.9$), whereas for orders deep in the queue ($M \approx 10{,}000$), it stops under only mildly favorable imbalance ($X \gtrsim 0.5$). Removing the deadline therefore does not eliminate time pressure so much as replace it with queue-position pressure: patience is high when the resting order is close to filling and falls off sharply as queue position worsens.

This is consistent with the execution-time distributions in Figure 23. The policy resolves the large majority of trades almost immediately (mode at $t \approx 0$–$5\,\mathrm{s}$) with a long right tail out to roughly 130 s, whereas the non-adaptive Always-LO benchmark spreads its execution mass more evenly over 0–50 s with a fatter tail extending to nearly 200 s, since it holds its position until forced out by a price move regardless of the imbalance signal.



(a) Policy execution time.

(b) Always-LO execution time.

Figure 23: Execution-time distributions: policy vs. Always-LO under $T = \infty$.

Table 4 reports the aggregate backtest statistics. The stationary policy achieves a mean PnL of $-0.258¢$ per trade (median $-0.50¢$), versus $-0.503¢$ for Always-MO and $-0.508¢$ for Always-LO – roughly a 49% reduction in mean cost relative to Always-MO. The policy submits a market order outright in 59.7% of trades, achieves a passive limit-order fill in 32.2%, and is forced out by an adverse price move in the remaining 8.1%; as expected, the deadline-triggered exit rate is exactly 0%, since no such deadline exists in this configuration.

Table 4: Stationary ($T = \infty$) policy backtest summary.

Metric	Value
Test windows	360
Policy mean PnL	$-0.258¢$
Policy median PnL	$-0.500¢$
Always-MO mean PnL	$-0.503¢$
Always-LO mean PnL	$-0.508¢$
Stop rate (MO submitted)	59.7%
LO fill rate	32.2%
Price-drop-triggered stop rate	8.1%
Deadline-triggered stop rate	0.0%
Mean execution time	8.12 s

4 Appendix

4.1 Estimating Probabilities

In order to ensure accuracy for our model, we needed to calculate sufficient sample sizes to estimate the transition probabilities for LO imbalance. In our initial model, we decided to have $k = 10$ imbalance states, namely $\{-0.9, -0.7, -0.5, -0.3, -0.1, 0.1, 0.3, 0.5, 0.7, 0.9\}$. Note that having the states range from -0.9 to 0.9 ensures that the intervals of values which round to each state all have the same width, which prevents extreme states from being automatically underestimated.

For each of our 10 states, we need to estimate the probability of the imbalance increasing or decreasing in the next time step. This means we need to estimate 20 parameters using our data.

Consider some imbalance value x_i. Let $\zeta_1, ..., \zeta_N$ be random variables representing the sub-

sample of times such that $I_t = x_i$. Assume we are trying to estimate $q_{up}(x)$, or in other words, $\mathbb{P}(I_{t+1} = x_{i+1} \mid I_t = x_i)$. We define the random variables by

$$\zeta_j = \begin{cases} 1 & \text{with probability } q_{up}(x_i) \\ 0 & \text{otherwise} \end{cases}$$

Then, we have that $\mathbb{E}(\zeta_j) = q_{up}(x_i)$. Now we can take $\overline{\zeta_N} = \frac{1}{N} \sum_{j=1}^{N} \zeta_j$, and since any ζ_j has the same expected value, we have that $\mathbb{E}(\overline{\zeta_N}) = \frac{1}{N} \cdot N \cdot \mathbb{E}(\zeta_j) = \mathbb{E}(\zeta_j) = q_{up}(x_i)$. By the Law of Large Numbers, we know that $\lim_{N\to\infty} \overline{\zeta_N} = \mathbb{E}(\zeta_j) = q_{up}(x_i)$.

Our sample sizes obviously must be finite, so we must find a size that makes us highly likely to have an accurate estimate. We first note that $\overline{\zeta_N} - \mathbb{E}(\zeta_j) = \frac{1}{N} \sum_{j=1}^{N} (\zeta_j - \mathbb{E}(\zeta_j))$. We define $\eta_N = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} (\zeta_j - \mathbb{E}(\zeta_j))$, which gives us $\overline{\zeta_N} - \mathbb{E}(\zeta_j) = \frac{1}{\sqrt{N}} \eta_N$.

By the Central Limit Theorem, as N increases, η_N becomes well estimated by Normal$(0, \sigma^2)$, where σ^2 respresents the variance of ζ_j, which is equal to $q_{up}(x_i)(1 - q_{up}(x_i))$, since each ζ_j is distributed Bernoulli$(q_{up}(x_i))$.

Now, if we divide by the standard deviation σ. we have $Z = \frac{1}{\sigma} \eta_N$, which is now distributed Normal(0, 1). Since $\sigma = \sqrt{q_{up}(x_i)(1 - q_{up}(x_i))}$, we get a final equation of $\overline{\zeta_N} - \mathbb{E}(\zeta_j) = \sqrt{\frac{q_{up}(x_i)(1-q_{up}(x_i))}{N}} \cdot Z$. The same theory holds for q_{down} and any other estimated probabilities. As all probabilities are unknown, we will use the sample value to compute the error.

It is a well-known fact that $\mathbb{P}(-1.96 < Z < 1.96) \approx 0.95$. Thus, for some given error ϵ, we can solve for an N that will give us 95% confidence of our true value falling within that error. It is also worth noting that we always have $q_{up}(x_i)(1 - q_{up}(x_i)) \leq 0.25$. Then,

$$\begin{aligned} \epsilon &= \sqrt{\frac{q_{up}(x_i)(1 - q_{up}(x_i))}{N}} \cdot Z \\ &\leq \sqrt{\frac{0.25}{N}} \cdot 1.96 \\ &= \frac{0.98}{\sqrt{N}} \\ \implies N &= \left(\frac{0.98}{\epsilon}\right)^2 \end{aligned}$$

is a conservative choice for N that will yield us at least 95% confidence for our given error. For the sake of our experiment, we felt that an estimate within 0.01 of the actual probability was sufficient to accurately model our problem. Setting $\epsilon = 0.01$ yields $N = 9604$. As seen in Figure 4, our data from November 21, 2014 for MSFT stock has a sufficient sample size for all k imbalance states.

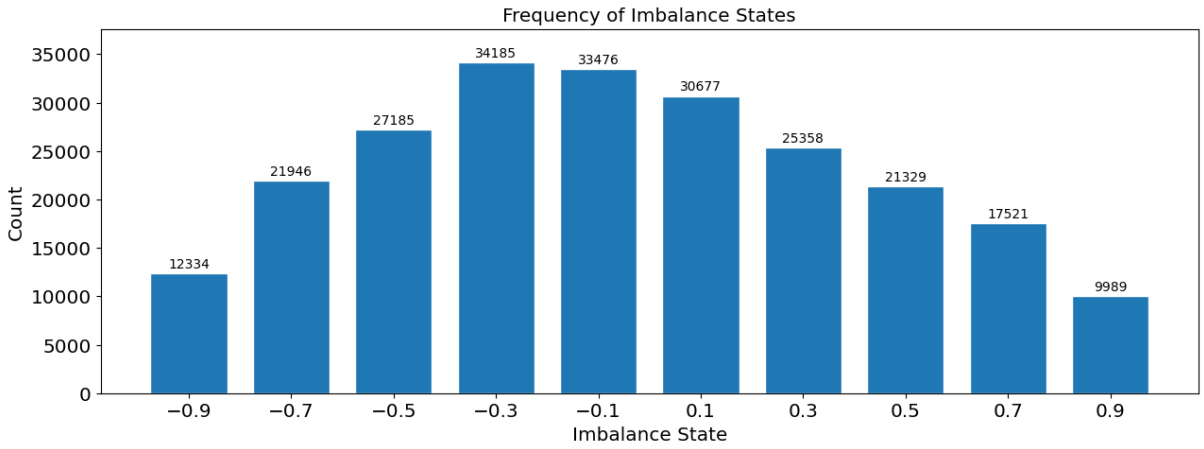


Figure 24: Imbalance state frequencies for MSFT stock (November 21, 2014)

4.2 Contraction of the Empirical Snell Operator

We now justify the convergence of the Snell operator used in the empirical test. Let k be the number of states and consider the space $\mathbb{R}^k$ equipped with the sup norm

$$\|u\|_\infty := \max_{1\le i\le k} |u_i|.$$

This is a complete metric space under the metric

$$d(u,w) := \|u-w\|_\infty.$$

For a value vector $u \in \mathbb{R}^k$, define the empirical Snell operator $A : \mathbb{R}^k \to \mathbb{R}^k$ componentwise by

$$(Au)_i = \max\left[-\frac{s}{2}, -\frac{3s}{2}p_i + \frac{s}{2}\pi_i + (1-p_i-\pi_i)\sum_{j=1}^{k} P_{ij}u_j\right],$$

where P_{ij} is the empirical transition matrix, p_i is the probability of an adverse downward midprice move, and π_i is the probability of a limit order fill.

Assume

$$0 \le p_i, \pi_i \le 1, \qquad p_i + \pi_i \le 1,$$

and define

$$\beta := \max_{1\le i\le k}(1-p_i-\pi_i).$$

If

$$\beta < 1,$$

then A is a contraction in the sup norm.

To prove this, let $u, w \in \mathbb{R}^k$. Define the continuation value

$$C_i(u) := -\frac{3s}{2}p_i + \frac{s}{2}\pi_i + (1-p_i-\pi_i)\sum_{j=1}^{k} P_{ij}u_j.$$

Then

$$(Au)_i = \max\left[-\frac{s}{2}, C_i(u)\right].$$

Since the map $x \mapsto \max(a, x)$ is 1-Lipschitz for fixed a,

$$|(Au)_i - (Aw)_i| \leq |C_i(u) - C_i(w)|.$$

Now,

$$C_i(u) - C_i(w) = (1 - p_i - \pi_i) \sum_{j=1}^{k} P_{ij}(u_j - w_j).$$

Therefore,

$$|C_i(u) - C_i(w)| \leq (1 - p_i - \pi_i) \sum_{j=1}^{k} P_{ij}|u_j - w_j|.$$

Since P is a transition matrix,

$$P_{ij} \geq 0, \qquad \sum_{j=1}^{k} P_{ij} = 1.$$

Hence,

$$|C_i(u) - C_i(w)| \leq (1 - p_i - \pi_i)\|u - w\|_\infty.$$

Taking the maximum over all states i, we obtain

$$\|A(u) - A(w)\|_\infty \leq \max_i (1 - p_i - \pi_i)\|u - w\|_\infty = \beta\|u - w\|_\infty.$$

Thus, if $\beta < 1$, A is a contraction.

By the Banach fixed point theorem, A has a unique fixed point $u^* \in \mathbb{R}^k$ satisfying

$$u^* = A(u^*).$$

Moreover, for any initial vector $u^{(0)} \in \mathbb{R}^k$, the sequence

$$u^{(n+1)} = A(u^{(n)})$$

converges to u^* in the sup norm:

$$\lim_{n \to \infty} \|u^{(n)} - u^*\|_\infty = 0.$$

The fixed point u^* can be interpreted as the stationary infinite-horizon value function associated with the empirical state dynamics. The corresponding stopping rule is obtained by comparing the immediate stopping payoff $-s/2$ with the continuation value:

$$-\frac{3s}{2} p_i + \frac{s}{2}\pi_i + (1 - p_i - \pi_i) \sum_{j=1}^{k} P_{ij} u_j^*.$$

Thus, state i is a stopping state if and only if

$$-\frac{s}{2} \geq -\frac{3s}{2} p_i + \frac{s}{2}\pi_i + (1 - p_i - \pi_i) \sum_{j=1}^{k} P_{ij} u_j^*.$$

4.3 Neural Network Model

To move beyond the empirical $k \times k$ transition matrix, we implemented a neural transition model in pairwise form, following the notation

$$f_\theta(x_i, x_j) \to \hat{q}(x_i, x_j) \in [0, 1], \qquad \inf_\theta \sum_n \left(f_\theta(x_i, x_j) - \hat{q}(x_i, x_j)\right)^2,$$

where $\hat{q}(x_i, x_j) = \text{count}(i \to j)/\text{count}(i)$ is the empirical transition frequency between discretized states i and j. The network takes the concatenated pair (x_i, x_j) as input and is trained by MSE regression against these empirical frequencies.

The intended advantage of this formulation is interpolation: once trained, f_θ can be queried on *any* pair (x_i, x_j), including state pairs that were never observed together in the training data (zero empirical count), giving those pairs a learned, non-zero transition probability instead of the hard zero an empirical frequency table would assign. Building the full transition matrix requires evaluating f_θ on every one of the K^2 possible state pairs:

This model was not used for the main results because the learned boundaries were not satisfactory. Figure 25 shows the stopping boundary produced by the pairwise model, one panel per queue-position bin. Across most bins the boundary sits flat near the lower imbalance limit for almost the entire 600-step horizon and only jumps sharply in the last ∼50 steps as the terminal reward begins to dominate. This is consistent with the MSE regression target being noisy: many (i, j) cells have small or zero transition counts, so the empirical frequencies $\hat{q}(x_i, x_j)$ used as training labels are themselves poorly estimated, and the pairwise network has no mechanism to distinguish a genuinely low transition probability from an under-sampled one.

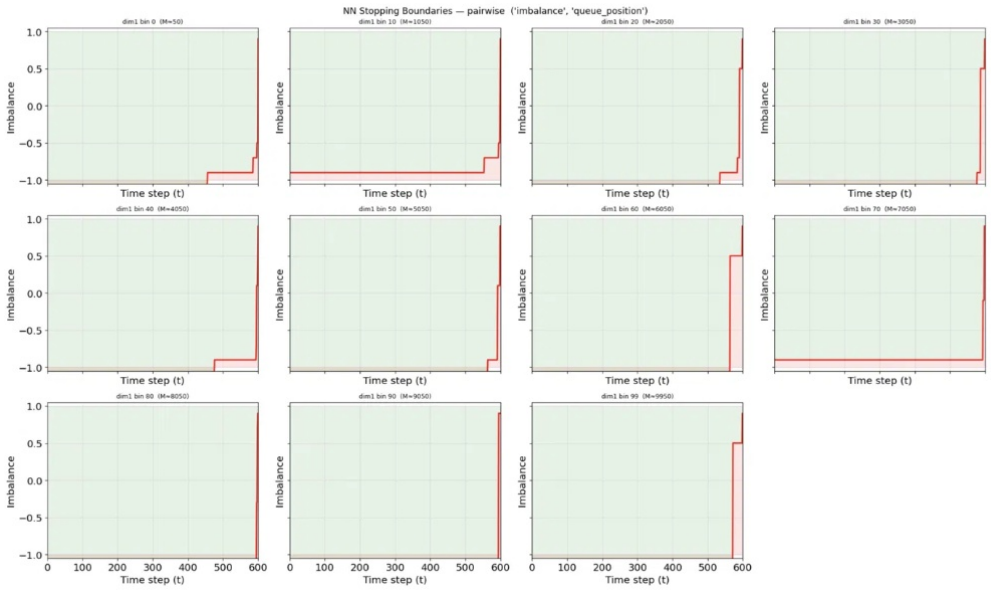


Figure 25: Stopping boundaries learned by the pairwise f_θ transition model, shown across queue-position bins. Boundaries are largely flat until late in the horizon rather than varying smoothly with time and state.

Given the unconvincing boundary quality and the limited time remaining in the project, the pairwise formulation was not pursued further.